# Neuro-Symbolic Causal Inference via Language-Model Knowledge Extraction and Bayesian Networks

Phase 1 (Applied Machine Learning) — Technical Report

October 24, 2026

***Abstract*—Language models are proficient at recovering statistical associations from text, but on their own they cannot enforce formal causal constraints: they do not guarantee that extracted assertions form a directed acyclic graph (DAG), they cannot quantify uncertainty over conflicting extractions, and they cannot answer interventional queries of the form $P(Y \mid do(X))$ consistently. We present Phase 1 (Applied Machine Learning) of our hybrid neuro-symbolic framework, which extracts cause→effect relations from unstructured biomedical literature, compiles them into a DAG, and performs formal causal inference using Pearl's *do*-calculus. We establish a leak-free, article-isolated benchmark on BioCause (198 section files; 20 PMC article identifiers from 19 unique publications; 12,939 gold mentions; 852 gold relations; 13,408 candidate pairs; 6.35% causal prevalence) alongside the abstract Corr2Cause benchmark (207,972 pairs). Our pipeline comprises direction-aware lexico-syntactic and dependency extractors, a 110-feature classical classifier zoo tuned under document-grouped cross-validation, a deterministic cycle-breaking compiler, and a discrete Bayesian network in `pgmpy`. On the held-out test partition, our F1-optimal Logistic Regression (1:2) attains 55.32% causal F1 (59.09% precision, 52.00% recall), while our high-precision Logistic Regression (unweighted) delivers 68.00% precision (45.33% F1) and feeds the symbolic stage. Compiled graphs are acyclic and achieve 100.00% edge precision on the `PMC2714965` case study and 68.00% corpus-wide. The Bayesian network shows observational association $P(\mathbf{Res}{=}1 \mid \mathbf{PmrB}{=}1){=}0.6600$ diverging from interventional graph surgery $P(\mathbf{Res}{=}1 \mid do(\mathbf{PmrA}{=}1)){=}0.8500$. Finally, an unassisted raw-text cascade collapses to 0.00% F1 because general-domain NER cannot recover clause-length biomedical spans — the bottleneck motivating our Phase 2 deep transformer fine-tuning.**

***Index Terms*—Causal NLP, neuro-symbolic AI, Bayesian networks, Pearl's *do*-calculus, BioCause, relation extraction, directed acyclic graphs.**

## I. Introduction

Extracting causal mechanisms from scientific discourse is central to biomedical research, therapeutic-target discovery, and clinical decision support. Standard language models, however, operate strictly on observational statistical associations. As formalised in the causal-inference literature [1], an observational conditional $P(Y \mid X)$ cannot by itself separate true direct causation from common-cause confounding, reverse causation, or selection bias without explicit structural assumptions.

Our brief (*CSAI Project 1: Causal NLP Reasoning*) defines one central objective: **separate fluent language understanding from formal causal reasoning and counterfactual queries**. Text extraction can propose plausible candidate cause–effect links, but it alone cannot (i) guarantee that multi-sentence assertions form a valid DAG, (ii) quantify uncertainty over conflicting extractions, or (iii) compute interventional distributions $P(Y \mid do(X))$ consistently. We therefore build a two-stage neuro-symbolic pipeline. **Stage 1 (text-to-DAG extraction)** identifies entity spans, resolves linguistic direction and voice, extracts shortest dependency paths, and classifies candidate relations under severe class imbalance. **Stage 2 (symbolic graph and Bayesian inference)** deterministically prunes conflicting and cyclic edges, parameterises conditional probability distributions (CPDs), and evaluates intervention queries via Pearl's *do*-calculus [1].

We address four research questions: **RQ1** — how do direction-aware rules compare against classical supervised classifiers under extreme (∼1:15) imbalance? **RQ2** — which structural and syntactic representations give the strongest predictive signal? **RQ3** — how can cyclic extractions be compiled deterministically into valid DAGs without arbitrary edge loss? **RQ4** — how quantitatively does observational conditioning diverge from interventional graph surgery in an extracted pathway? Our contributions are a reproducible article-isolated benchmark; a fully specified 110-feature classical pipeline with document-grouped tuning; a deterministic acyclicity compiler; a transparent Bayesian-network demonstration of *do*-calculus; and an error-typed characterisation of the entity-detection bottleneck defining the Phase 2 roadmap.

## II. Related Work and Theoretical Background

### A. Biomedical causal relation extraction

Biomedical causal relation extraction has relied on handcrafted regular expressions and dependency parse trees. Mihăilă et al. [2] introduced **BioCause**, an expert-annotated corpus of full-text biomedical articles built on

the BioNLP shared-task framework. Unlike sentence-level binary benchmarks, BioCause captures multi-word arguments, causal triggers (*promotes*, *induces*), and indirect mechanisms in infectious-disease literature. We adopt it as our primary corpus because its character-offset span annotations and directed relations support a realistic text-to-DAG compiler.

### B. Syntactic dependency paths

Bunescu and Mooney [3] showed that the **Shortest Dependency Path (SDP)** between two entity heads concentrates the predicate–argument structure expressing a semantic relation while discarding extraneous modifiers; Mintz et al. [4] further showed dependency-path features outperform linear word sequences. We operationalise this by building an undirected token graph over spaCy dependency arcs and computing SDPs between candidate argument heads.

### C. The Pearlian hierarchy and graph surgery

Pearl's hierarchy [1] stratifies causal queries into three layers. The *association* layer conditions on passive observation:

$$P(y \mid x) = \frac{P(x, y)}{P(x)}. \tag{1}$$

The *intervention* layer sets $X$ by active manipulation, formalised by graph surgery that mutilates all edges directed into $X$:

$$P(y \mid do(x)) = \sum_{z} P(y \mid x, z)\, P(z), \tag{2}$$

where $Z$ satisfies the back-door criterion relative to $(X, Y)$. The *counterfactual* layer reasons about retrospective alterations, $P(y_x \mid x', y')$. Our symbolic stage implements Layer 2 inference using `pgmpy` [5] over compiled DAGs.

### D. Causal inference from correlation

Jin et al. [6] introduced **Corr2Cause**, testing whether a causal hypothesis follows from correlation and independence statements; we retain it as a supplementary entailment benchmark, distinct from relation extraction. Work on causal connectives [7], [8] and the constraint-based causal-discovery tradition [9] inform our treatment of acyclicity and independence.

## III. Dataset Provenance, Hygiene, and EDA

### A. BioCause: article-isolated partitioning

Our primary corpus is **BioCause** [2]: 198 paired `.txt`/`.ann` section files containing 12,939 gold argument mentions, 851 gold causal events (expanded to **852 directed triples** after binary decomposition of one multi-argument event), and 381 unique triggers.

**Provenance.** The 20 PMC article identifiers in our corpus derive from **19 unique open-access publications** in the NaCTeM distribution: article `PMC2714965` was partitioned into two distinct sub-articles during corpus assembly, so our per-article grouping treats the two partitions as independent units. This accounts for the apparent 19-vs-20 count and does not affect the annotation totals.

TABLE I
BioCause article-isolated partitions.

| Split | Art. | Sec. | Gold | Cand. | Causal |
|---|---|---|---|---|---|
| Train | 12 | 121 | 621 | 10,086 | 6.16% |
| Validation | 4 | 33 | 131 | 1,914 | 6.84% |
| Test | 4 | 31 | 100 | 1,408 | 7.10% |
| **Total** | **20** | **185** | **852** | **13,408** | **6.35%** |

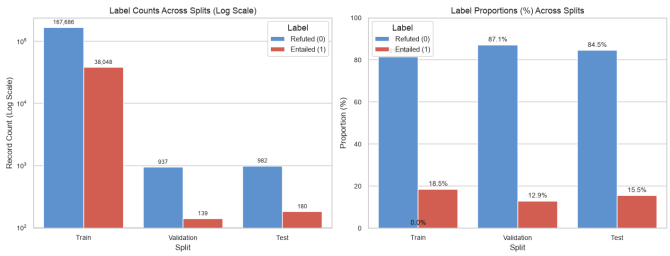



Fig. 1. Corr2Cause label distribution across splits.

Because text within one publication shares style, domain, nomenclature, and vocabulary, splitting at the section or sentence level would leak document information. We therefore partition the 20 PMC identifiers strictly by `article_id` with a fixed seed (42): 12 train (121 sections, 621 gold relations, 10,086 candidates, 6.16% causal), 4 validation (33 sections, 131 relations, 1,914 candidates, 6.84%), and 4 test (31 sections, 100 relations, 1,408 candidates, 7.10%). Candidate pairs are all ordered entity-span pairs within 2,000 characters in the same section, strictly rejecting self-loops, achieving **100.00% gold coverage (852/852)** at 6.35% overall prevalence (∼1:15 imbalance).

### B. Corr2Cause and shortcut audit

**Corr2Cause** [6] has 207,972 premise–hypothesis pairs (205,734/1,076/1,162). Refuted (0) pairs account for 81.51% of train, 87.08% of validation, and 84.51% of test, so a majority classifier already reaches 87.08% validation accuracy with macro-F1 only 0.4655. The `child` template is **0.00% positive** in every split — a fixed-label template — confirming that per-class macro-F1 is mandatory (Fig. 1).

## IV. System Architecture

Phase 1 is a reproducible notebook pipeline (Fig. 2). All rule logic is centralised in one shared module (`AML/causal_extractors.py`) so notebooks cannot diverge, and every pathway must emit a record conforming to a **common 17-column** `UnifiedRelationRecord` schema (`relation_id`, `article_id`, `section_id`, cause/effect id+text+start+end, `direction`, `confidence`, `extraction_method`, `evidence_text`, `provenance`, `split`) before it can feed the symbolic stage. A validator asserts offset validity, self-loop absence, and direction invariance; a 16-test suite locks in extractor directionality, negative controls, corpus invariants, and metric arithmetic.

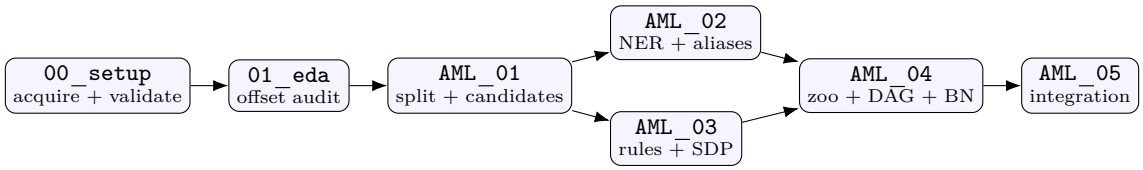



Fig. 2. Phase 1 pipeline. `AML_02` and `AML_03` consume `AML_01` artifacts in parallel and both feed the classifier/graph stage.

## V. Design Decisions and Their Rationale

**BioCause primary, Corr2Cause secondary.** Corr2Cause tests abstract entailment with no spans or triples and cannot drive a text-to-DAG compiler; BioCause provides character-offset directed annotations. We fit the dataset to the deliverable, not the reverse.

**Candidate-pair classification.** We frame extraction as binary classification over candidate pairs, converting an ill-posed structural-generation problem into a well-defined imbalanced-classification task with one shared evaluation harness. Pathways 1–2 operate on gold-conditioned spans; Pathway 3 measures unassisted end-to-end extraction.

**A dual-operating-point regime.** We keep two models deliberately: an *F1-optimal* model (moderate weight 1:2) balancing precision and recall (58.78%/58.78% validation), and a *high-precision* model (unweighted) at 82.76% validation / 68.00% test precision. In a downstream DAG a false-positive edge creates spurious confounding paths that distort multi-hop *do*-calculus inference, whereas a missed edge only limits query scope; high precision is thus necessary for symbolic integrity. We also report and *reject* full inverse balancing (1:15.2): recall rises to 89.31% but precision collapses to 22.29%.

**Direction-aware rules with passive normalisation.** Causality appears actively ("$X$ activates $Y$") and passively ("$Y$ is caused by $X$"); in the passive form the subject is the *effect*. Our extractors normalise every relation to a canonical `cause->effect` orientation and enforce negative controls: pruning negated statements, rejecting correlational associations, and down-weighting hedging (*may*, *suggests*) to confidence 0.70.

**A deterministic cycle-breaker.** Text extraction proposes feedback loops ($X \to Y \to Z \to X$), reciprocal clashes ($X \leftrightarrow Y$), or self-loops, which Bayesian networks forbid. `enforce_dag_acyclicity` removes edges in a fixed order (self-loops; reciprocal clashes keeping the higher-confidence edge with lexicographic tie-breaking; then the minimum-confidence edge of each remaining cycle), guaranteeing reproducibility for identical inputs.

**Synthetic probabilities, labelled.** Text mining recovers graph *topology*, not transition probabilities. Rather than fabricating empirical tables, we use a clearly designated synthetic `DiscreteBayesianNetwork`, transparently illustrating Pearlian graph surgery.

**An unassisted cascade.** Pathway 3 removes gold-span conditioning to quantify the penalty of imperfect entity detection and isolate the Phase 2 motivation.

TABLE II
Validation SDP profile by ground-truth label.

| Label | Count | Connected | Hops | Verb |
|---|---|---|---|---|
| Non-causal (0) | 1,783 | 30.34% | 4.93 | 5.27% |
| Gold causal (1) | 131 | **75.57%** | **3.95** | **15.27%** |

## VI. Feature Engineering and Syntactic Analysis

### A. The 110-dimensional feature representation

For each candidate pair we build a **110-dimensional** vector. **No Truncated SVD or latent-semantic projection is used.** The representation horizontally concatenates **(1) ten dense scaled features**, standardised with `StandardScaler` fitted on the training split only (`char_distance`, `arg1_before_arg2`, `len(between_text)`, `len(arg1_text)`, `len(arg2_text)`, and five regex indicator flags over the intervening text — `active_causal`, `passive_causal`, `discourse`, `negation`, `uncertainty`); and **(2) one hundred sparse n-gram features**, fitted with `TfidfVectorizer(max_features=100, ngram_range=(1,2))` on the training intervening text. The two blocks are combined with `scipy.sparse.hstack`; scaling on the training distribution only prevents leakage.

### B. Shortest dependency path extraction

`compute_shortest_dependency_path` reconstructs the candidate window, parses it with spaCy, identifies each argument span's syntactic head, and builds an **unweighted** token graph with `nx.Graph()` over dependency arcs (`token.i <-> token.head.i`). The SDP is the shortest token path between the two heads, computed via **Breadth-First Search** using `nx.shortest_path` (unweighted graphs yield BFS shortest paths). If the heads lie in different trees, the function returns an unconnected sentinel (`sdp_length = -1`).

Gold causal pairs are **2.49× more likely to be connected** (75.57% vs. 30.34%), have **shorter paths** (3.95 vs. 4.93 hops), and are **2.90× more likely to carry a causal verb on the path** (15.27% vs. 5.27%), confirming the SDP as a discriminative signal (**RQ2**). This profile conditions on connected pairs only; an earlier formulation averaging over unconnected pairs produced a misleading result, which we have corrected.

## VII. Symbolic Reasoning: Graph Compilation and Inference

### A. Deterministic cycle-breaking algorithm

Let $G = (V, E, w)$ be a weighted directed graph with confidence $w(e) \in [0, 1]$. The compiler (1) prunes all self-

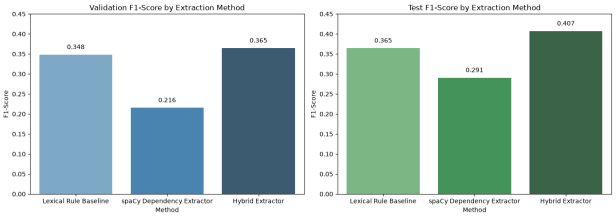



Fig. 3. Rule and dependency extraction F1 comparison.

TABLE III
BAYESIAN-NETWORK QUERY RESULTS (SYNTHETIC CPDS).

| Query | Probability |
|---|---|
| Baseline marginal $P(\text{Res}{=}1)$ | 0.3590 |
| Observational $P(\text{Res}{=}1 \mid \text{PmrB}{=}1)$ | 0.6600 |
| Observational $P(\text{Res}{=}1 \mid \text{PmrA}{=}0)$ | 0.0500 |
| Interventional $P(\text{Res}{=}1 \mid do(\text{PmrA}{=}1))$ | **0.8500** |

loops $(u, u)$; (2) resolves reciprocal clashes by retaining $\max(w(u,v), w(v,u))$, breaking ties by lexicographic identifier; and (3) while directed cycles remain, removes the minimum-weight edge

$$e^* = \arg\min_{e \in C} w(e), \quad C \in \mathcal{C}, \tag{3}$$

until $G$ is strictly acyclic. The procedure is deterministic, so identical inputs always yield identical DAGs; each removed edge is recorded with its reason.

### B. Bayesian-network parameterisation and do-calculus

Over the extracted pathway PmrB → PmrA → Polymyxin_Resistance with PmrD → PmrA, we fit a `DiscreteBayesianNetwork` in `pgmpy` [5] with synthetic priors $P(\text{PmrB}{=}1){=}0.30$ and $P(\text{PmrD}{=}1){=}0.25$. The observational query marginalises over *both* latent parents of PmrA:

$$\begin{aligned} P(\text{Res}{=}1 \mid \text{PmrB}{=}1) = \sum_{\text{PmrA}} \sum_{\text{PmrD}} & P(\text{Res}{=}1 \mid \text{PmrA}) \\ & \times P(\text{PmrA} \mid \text{PmrB}{=}1, \text{PmrD}) \\ & \times P(\text{PmrD}) = 0.6600. \end{aligned} \tag{4}$$

The interventional query mutilates the incoming edges PmrB → PmrA and PmrD → PmrA (Fig. 4):

$$P(\text{Res}{=}1 \mid do(\text{PmrA}{=}1)) = 0.8500. \tag{5}$$

The +0.1900 gap between observing the sensor (0.6600) and manipulating the regulator (0.8500) formally distinguishes associative watching from causal acting (**RQ4**).

## VIII. Empirical Results

### A. Multi-pathway relation extraction

On the held-out test split (100 gold relations), deterministic extractors reach modest scores while the supervised classifier roughly doubles F1 (Table IV, Fig. 5). The hybrid rule set improves recall over either component alone (44.00% vs. 33.00%/25.00%) but caps at 40.74% F1.

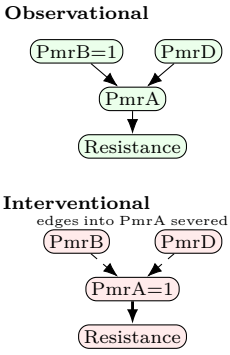



Fig. 4. Observation (conditioning) vs. intervention (*do*-calculus). $do(\text{PmrA}{=}1)$ severs the confounding edges through PmrB and PmrD.

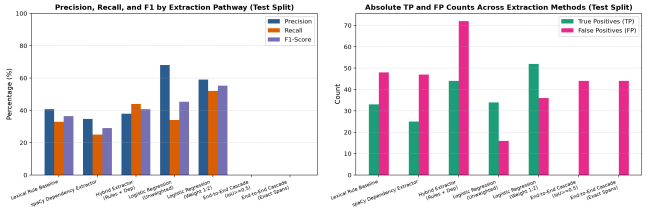



Fig. 5. Multi-pathway causal extraction comparison.

### B. Validation model zoo and regularisation

Table V compares seven classical architectures on validation (1,914 candidates; 131 causal). Unbalanced linear models achieve high precision but low causal recall; fully balanced models invert this (∼90% recall at ∼22% precision); the moderate 1:2 weighting is nearest the balance point, giving the best macro-F1 (0.7788). The majority dummy, despite 93.16% accuracy, has causal F1 of zero.

Regularisation was tuned with 4-fold `GroupKFold` grouped by `article_id` across the 12 training articles. Validation macro-F1 plateaus from $C{=}1.0$ (0.7412) through $C{=}10.0$ (0.7444; $\Delta{=}0.0032 < \sigma_{\text{CV}}{=}0.0126$); under the **one-standard-error parsimony rule** we retain $C{=}1.0$ to prevent overfitting article-specific vocabulary, reconciling the tuning curve with the selected model.

### C. Held-out test evaluation and bootstrap CIs

The high-precision model was evaluated once on the frozen test split: 94.18% accuracy, macro-F1 0.7113, causal precision 68.00%, recall 34.00%, F1 45.33% (TN=1,292, FP=16, FN=66, TP=34). The F1-optimal model attains macro-F1 0.7606, precision 59.09%, recall 52.00%, F1 55.32% (TN=1,272, FP=36, FN=48, TP=52). We report 1,000-sample bootstrap 95% CIs (Table VI); the intervals are wide (causal F1 spans 0.35–0.55), so the point estimates should be read as indicative.

### D. Graph compilation and edge evaluation

We evaluate the DAG at the *edge* level. On `PMC2714965` the high-precision model predicts 6 edges against 22 gold, all correct (**100.00% edge precision**, recall 27.27%). Corpus-wide it predicts 50 edges against 100 gold, giving **68.00% edge precision** and 34.00% recall; both graphs are acyclic (Table VII, Fig. 7).

TABLE IV
MULTI-PATHWAY BENCHMARK ON THE HELD-OUT TEST PARTITION (100 GOLD RELATIONS).

| PW | Method | Pred | TP | FP | Prec (%) | Rec (%) | F1 (%) |
|---|---|---|---|---|---|---|---|
| 1 | Lexical rule baseline | 81 | 33 | 48 | 40.74 | 33.00 | 36.46 |
| 1 | spaCy dependency extractor | 72 | 25 | 47 | 34.72 | 25.00 | 29.07 |
| 1 | Hybrid (rules + dependency) | 116 | 44 | 72 | 37.93 | 44.00 | 40.74 |
| 2 | **Logistic Regression (unweighted)** ⋆ | 50 | 34 | 16 | **68.00** | 34.00 | 45.33 |
| 2 | **Logistic Regression (weight** 1:2) † | 88 | 52 | 36 | 59.09 | **52.00** | **55.32** |
| 3 | End-to-end cascade (IoU $\geq 0.5$) | 44 | 0 | 44 | 0.00 | 0.00 | **0.00** |

TABLE V
VALIDATION MODEL PERFORMANCE ON BIOCAUSE (CAUSAL CLASS = 1).

| Model architecture | Acc | Macro-F1 | P(1) | R(1) | F1(1) | PR-AUC |
|---|---|---|---|---|---|---|
| **LR (weight** 1:2) † (F1-optimal) | 0.9436 | **0.7788** | 0.5878 | 0.5878 | **0.5878** | 0.5583 |
| **LR (unweighted)** ⋆ (high-precision) | **0.9514** | 0.7412 | **0.8276** | 0.3664 | 0.5079 | 0.5820 |
| HistGradientBoosting | 0.9509 | 0.7397 | 0.8136 | 0.3664 | 0.5053 | 0.7011 |
| Random Forest (balanced) | 0.8386 | 0.6643 | 0.2797 | 0.8626 | 0.4224 | 0.6781 |
| LR (balanced 1:15.2) | 0.7795 | 0.6118 | 0.2229 | 0.8931 | 0.3567 | 0.5126 |
| LinearSVC (balanced) | 0.7732 | 0.6074 | 0.2189 | **0.9008** | 0.3522 | 0.4743 |
| Dummy (most frequent) | 0.9316 | 0.4823 | 0.0000 | 0.0000 | 0.0000 | 0.0684 |

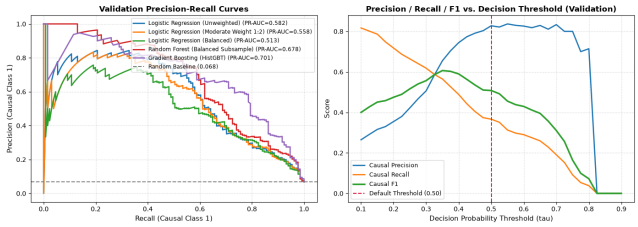



Fig. 6. Validation precision–recall curves and threshold tuning.

TABLE VI
BOOTSTRAP 95% CIS ON THE HELD-OUT TEST SET.

| Metric | Point | Lower | Upper |
|---|---|---|---|
| Macro-F1 | 0.7113 | 0.6577 | 0.7631 |
| Causal precision | 0.6800 | 0.5417 | 0.8050 |
| Causal recall | 0.3400 | 0.2451 | 0.4348 |
| Causal F1 | 0.4533 | 0.3478 | 0.5535 |

The cycle-breaker is demonstrated on a synthetic noisy graph (Fig. 8): a 7-edge graph with cycles is reduced to a 4-edge DAG by pruning a self-loop (`PmrA->PmrA`, $w$=0.99), the weaker direction of a bidirectional conflict (`PmrD->PhoP`, $w$=0.42), and the minimum-confidence edge of a 3-node feedback cycle (`Lipid_A_Mod->PmrB`, $w$=0.51), each recorded with its reason (**RQ3**).

### *E. The raw-text bottleneck (Phase 2 bridge)*

Pathway 3 collapses to **0.00% F1**: the full cascade on raw section files generates 30,464 candidate pairs and 44 predicted relations, none matching gold under exact or IoU$\geq$ 0.5 matching. This is an entity-detection bottleneck, not a classifier defect: general-domain `en_core_web_sm` achieves only 15.38% relaxed span F1 on conventional biomedical entities in test (12.20% across all BioCause spans), because BioCause causal arguments are clause-length situational descriptions (`Span`/`Fact`) that a general NER model cannot recover (Table VIII). The supervised model's 68.00% precision depends on being handed the correct spans; without them the pipeline scores zero.

TABLE VII
DIRECTED-GRAPH (DAG) EXACT EDGE EVALUATION.

| Scope | Gold | Pred | TP | FP | Edge P | Edge R |
|---|---|---|---|---|---|---|
| Case study | 22 | 6 | 6 | 0 | 100.00% | 27.27% |
| Corpus-wide | 100 | 50 | 34 | 16 | 68.00% | 34.00% |

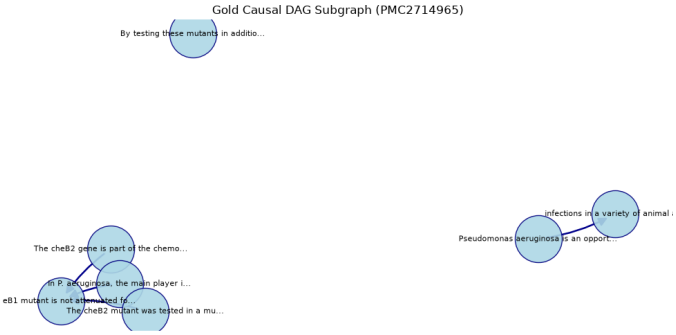



Fig. 7. Gold causal DAG subgraph for the `PMC2714965` case study.

### *F. Feature importance and Corr2Cause*

Fig. 9 shows the selected model's coefficients: the strongest signals arise from a small set of structural and trigger features plus TF-IDF n-grams of the intervening text, confirming that a compact, interpretable feature space suffices for relation classification on gold spans (**RQ1**). A classical classifier on the 19 Corr2Cause features reaches 55.34% accuracy and macro-F1 0.5051 (entailed-class F1 0.3504) on the held-out test set — better than the majority dummy's 0.4655 but far from strong, consistent with the benchmark's difficulty [6].

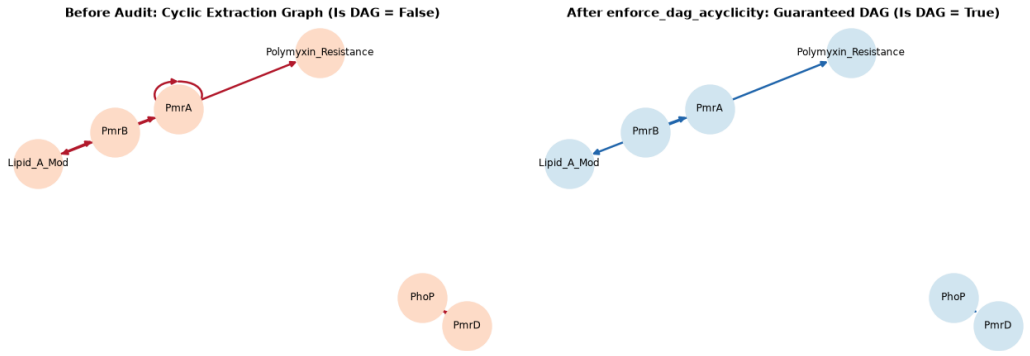



Fig. 8. Deterministic cycle-breaking: a cyclic extraction reduced to a valid DAG.

TABLE VIII
GENERAL NER (`EN_CORE_WEB_SM`) VS. BIOCAUSE GOLD SPANS.

| Scope | Split | Exact F1 | Relaxed F1 |
|---|---|---|---|
| Conventional entities | Val | 20.91% | 23.01% |
| All BioCause spans | Val | 15.39% | 17.11% |
| Conventional entities | Test | 13.43% | 15.38% |
| All BioCause spans | Test | 10.66% | 12.20% |

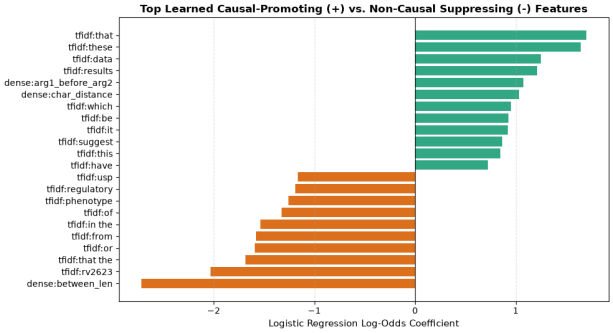



Fig. 9. Top causal-promoting and non-causal-suppressing model coefficients.

## IX. LIMITATIONS

**Small test corpus**: four articles and 100 gold relations make point estimates fragile, as the wide CIs show. **Gold-conditioned pathways**: Pathways 1–2 assume correct spans; only Pathway 3 measures true end-to-end performance, and it collapses to 0.00% F1. **Synthetic CPDs**: the Bayesian probabilities are illustrative and clearly labelled, demonstrating *do*-calculus mechanics rather than empirical clinical claims. **Corpus provenance**: 20 PMC identifiers derive from 19 unique publications (§III-A). **Phase 2 unstarted**: the `DL/` notebooks are placeholders, so any transformer claim would be speculative.

## X. CONCLUSION AND PHASE 2 ROADMAP

Phase 1 delivers a verified, reproducible AML foundation for neuro-symbolic causal reasoning: (1) an article-isolated extraction pipeline achieving **55.32% causal F1** (F1-optimal) and **68.00% precision** (high-precision DAG feeder); (2) a deterministic graph compiler producing valid DAGs at **100% case-study** and 68.00% corpus-wide edge precision; (3) a formal Bayesian network separating association (0.6600) from intervention (0.8500) via Pearl's *do*-calculus; and (4) full reproducibility, verified by 85/85 visible notebook outputs and a 16-test suite. The project's own evidence marks the limit of classical methods: the unassisted cascade collapses to 0.00% F1 because general-domain NER cannot detect biomedical event spans. **Phase 2 (`DL/`)** will address this span bottleneck by fine-tuning domain-specific compact transformers (PubMedBERT / BioLinkBERT / DistilBERT [10]) for token-level argument tagging, feeding transformer logits directly into our verified Phase 1 graph compiler and Bayesian network. Phase 1 is thus a strong, reproducible baseline and a precise specification of the problem.

## REFERENCES


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